ARTYOM HAYRAPETYAN

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Education

Yerevan State University Yerevan, Armenia

M.S. Mathematical & Software Development of Computing Systems and Networks Expected May 2027

B.S. Informatics and Applied Mathematics, Specialization in Information Security (GPA 16.7/20) May 2025

Experience

Polixis Geneva, Switzerland

Software Engineer 2024 – Present

- Designed and built a multi-tenant identity & access management platform from scratch (Java 25, Quarkus, Keycloak, PostgreSQL, jOOQ): raising throughput from ~600 to 3500+ requests/sec and cutting memory use by up to 30%.
- Engineered a crash-safe saga engine that keeps Keycloak and PostgreSQL consistent across multi-step operations, persisting every step and automatically compensating completed ones after a failure or restart.
- Built a data versioning & matching engine that merges each new dataset release into a master dataset of millions of records, cutting search latency by up to 70% with a custom in-memory B+ tree index (replacing MongoDB index lookups) and parallel processing.
- Built a resumable real-time sync service streaming data over compressed gRPC into MongoDB and Elasticsearch with parallel batching and checkpointing, deployed to multiple on-premise customer environments.
- Added reliable Kafka event publishing via the transactional outbox, load-tested services with k6, profiled them with JFR, and shipped every service to Kubernetes with Helm and GraalVM native builds.

Picsart Academy Yerevan, Armenia

Software Engineer / Practice Lead 2023 – 2024

- Developed and maintained applications in Java and Node.js, leading student teams through real-world projects from design and code review to delivery.
- Mentored developers on algorithms, data structures, problem solving and clean code.

European University of Armenia Yerevan, Armenia

Java Lecturer 2022 – 2023

- Taught Java and C++ to undergraduates and designed hands-on lab sessions for Java development.

UpWork Remote

Software Engineer (Freelance) 2018 – 2019

- Fixed and optimized backend systems; delivered small projects involving search, Kafka and Docker.

Projects

Armenian Voice AI | Real-Time Banking Voice Assistant Python, LiveKit, Claude, pgvector, React
github.com/Artyom7577/voice-ai

- Built an interruptible speech-to-text → Claude → text-to-speech pipeline in Armenian, a language most voice AI does not support, grounded by RAG over a pgvector knowledge base of bank documents.

Spring AI + RAG | LLM Features in Java Backends Java 21, Spring AI, Milvus
github.com/Artyom7577/spring-ai

- Ingested files and web pages into Milvus with chunking and embeddings, answering only from the top retrieved chunks and mapping LLM output to typed Java records.

Supermarket Voronoi Map | Bachelor's Thesis Java 21, Spring WebFlux, MongoDB
github.com/Artyom7577/undergraduate-project

- Collected supermarket locations across Yerevan via the Google Places API and computed a Voronoi diagram on a fully reactive stack to visualize each store's catchment area and underserved districts.

Skills

Languages: Java, Python, JavaScript, SQL, NoSQL, Bash

Backend: Spring Boot, Quarkus, Spring WebFlux, Mutiny, REST, gRPC, jOOQ, Flyway, Keycloak / OIDC

Data & Messaging: PostgreSQL, MongoDB, Redis, Apache Kafka, Milvus, pgvector

AI: LLMs, RAG, Spring AI, Claude API, Model Context Protocol (MCP), AWS

Infrastructure: Docker, Kubernetes, Helm, GraalVM, Prometheus, k6, JFR, Git, CI/CD, AWS

Certifications: AWS Developer – Associate (in progress) • Anthropic — Building with the Claude API · Claude Code

Spoken Languages: Armenian (Native), English, Russian